

# Real-time phase tracking for wide-band optical frequency measurements at the 20th decimal place

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**Optical frequency measurements are among the most precise tools available to science. With the rapid advances in optical clocks now achieving a low  $10^{-17}$  stability at 1 s and averaging down to the  $10^{-19}$  level in a few hundred seconds, real-time sensing of subtle phenomena becomes essential. To render possible such measurements, we introduce real-time optical phase tracking with ultra-low-noise frequency combs as a fundamental means to constantly monitor frequency offsets. This enables the characterization of optical frequency synthesis with stability and accuracy at the 20th decimal place within a measurement time of  $<100$  s. To enable comb operation at this level of performance, dichroic heterodyne detection is used to compensate phase drifts occurring in the generation and dissemination of widely spaced optical frequencies. We qualify an example set-up by comparison with a reference system, measuring an offset between two combs of  $(5.4 \pm 5.3) \times 10^{-21}$  in one single measurement run of  $1 \times 10^5$  s.**

Many applications take advantage of optical frequency combs<sup>1–5</sup> (FCs), including direct-, dual- and tri-comb spectroscopy<sup>6–11</sup>, attosecond physics<sup>12</sup>, photonics microwave generation<sup>13–15</sup> and accurate distance measurements<sup>16</sup>. Recently, FCs have even attained a new level of technological readiness for operations in space<sup>17</sup>. When employed as clockwork<sup>18,19</sup>, FCs must keep pace with the ever-improving stability and accuracy of optical frequency standards<sup>20–22</sup>, which rely on transition frequencies of hundreds of terahertz, now determined down to a relative accuracy to the 18th digit<sup>23–26</sup> and beyond<sup>27</sup>.

Research on optical clocks based on ultra-stable lasers<sup>28–31</sup> (USLs) and three-dimensional (3D) optical lattices<sup>21</sup> is making rapid progress towards achieving stabilities as low as  $10^{-17}$  at 1 s, averaging down to low  $10^{-19}$  values at 10,000 s (ref. 32). This improvement represents a level where clocks can become an intriguing tool for sub-centimetre measurement of the gravitational potential<sup>33,34</sup>. Quantification of the gravitational potential at these levels paves the way to explore new physics. The possible variation of fundamental constants and the plausibility of physical theories that go beyond the standard model are assessable via frequency ratio measurements between optical frequency standards of a different nature<sup>35,36</sup> at the aforementioned level of precision. As FCs phase-coherently link the different optical frequencies involved in these measurements, there is a major drive to improve their performance.

To measure relevant systematics at this level, the bridging FC must allow comparisons within short timescales, ideally as fast as the optical clocks permit. Here, we introduce two innovative concepts to ensure stable and accurate optical frequency measurements at levels comparable and beyond the best results reported so far. We first illustrate the analysis of phase drifts in real time for ultra-low-noise FCs, applicable to any electromagnetic frequencies. In particular, we use this technique to qualify frequency measurements with ratios up to  $\sim 1.45$ , bridging optical gaps up to  $\sim 90$  THz.

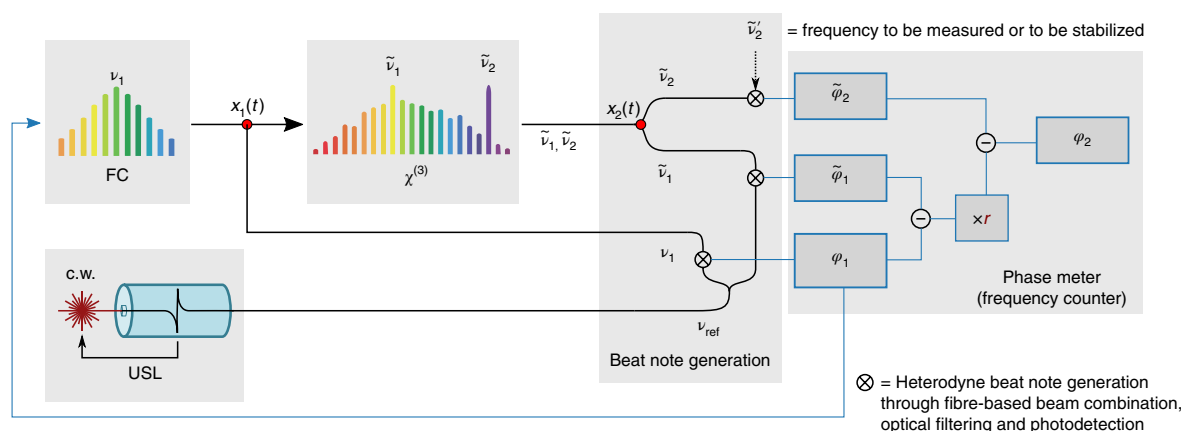
Second, we introduce passive dichroic heterodyne detection, an innovative concept to compensate such phase drifts, which allows us to demonstrate unprecedented agreement between two combs in the  $10^{-20}$  range within only 100 s. Together, these two techniques ensure that the key tool for precise optical frequency measurements introduces an error that is more than 100-fold lower than any optical clock uncertainty budget demonstrated so far.

Besides providing a direct link from the radio to the optical domain, FCs are the tool of choice to flexibly bridge the gap across the optical frequency domain for performing ratio measurements. Therefore, the true figure of merit for the qualification of FCs as a measurement tool is the achievable stability and accuracy with respect to the optical gap that is effectively bridged by the comb. The larger the gap, the larger the susceptibility to fundamental effects such as the unevenness of comb lines spacing over the spectrum on one hand, and the influence of technical noise of the FC's repetition rate on the other hand. Although there are now a few investigations of fractional instability or uncertainty at the level of  $10^{-19}$  and even below<sup>37–43</sup>, we currently see only one assessment in this range where the FC is truly qualified for bridging an optical gap<sup>39</sup>, namely of 23 THz. In that work, the transfer oscillator scheme<sup>44</sup> is used to strip off technical noise from the FC, and an ensemble of  $\sim 90$  measurement campaigns over the course of two weeks is used to obtain a fractional frequency offset of  $(-0.4 \pm 2.7) \times 10^{-21}$  between two combs of different embodiments (erbium fibre laser-based and Ti:sapphire laser-based FCs).

## Optical phase tracking and dichroic heterodyne detection

Here, we significantly push the limits of such frequency measurements, as we demonstrate similar performance in a much reduced acquisition time, while simultaneously dropping the need for the technically complex transfer oscillator scheme. Phase-drift analysis is performed on comb teeth over a broad range of frequencies, involving significant spectral broadening. As the optical frequency

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**Fig. 1 | Schematic of dichroic heterodyne detection for phase drift compensation.** A femtosecond FC seeds a spectral broadening unit, where the fibre links are indicated with black lines and radio-frequency signals with blue lines. In this unit, nonlinear  $\chi^{(3)}$  processes such as four-wave mixing or Raman shifting produce comb lines at spectral regions of interest, here indicated as  $\tilde{\nu}_2$  (the tilde indicates light affected by phase noise). Before entering the unit, the seeding light is split into two paths. One ( $\nu_1$ , propagating in a short passive fibre) is used to generate a beat note with a continuous-wave (c.w.) reference light  $\nu_{\text{ref}}$ , depicted as locked to a Fabry-Pérot cavity for obtaining a USL. Such a beat, containing phase reference information, is used to discipline the comb oscillator FC. The second split output, seeding the broadening unit, takes a long active and passive fibre optical path  $x_2(t) - x_1(t)$  until it reaches a dichroic splitter, accumulating short- and long-term phase noise. The fundamental light  $\tilde{\nu}_1$ , also affected by such noise, is beaten against the reference laser for generating a phase correction ( $\tilde{\varphi}_1 - \varphi_1$ ), while the nonlinearly shifted light  $\tilde{\nu}_2$  is used for measuring or stabilizing an external optical frequency  $\tilde{\nu}_2$ , carrying the phase information  $\tilde{\varphi}_2$ , affected by phase noise. Deducing experimentally the correcting factor  $r$ , a Doppler-compensated  $\varphi_2 = \tilde{\varphi}_2 - r(\tilde{\varphi}_1 - \varphi_1)$  phase is obtained, which up to small dispersive effects represents the ratio of the optical frequencies in the two heterodyne detections.

ratio to be assessed departs further from unity, separate branches, consisting of amplifiers and nonlinear optical processing seeded by the comb generator, are necessary to synthesize frequencies with high signal-to-noise ratio at the spectral regions of interest. The latter reduces the white phase noise plateau of the heterodyne beat notes, permitting higher precision to be reached in a shorter averaging time. While facilitating the analysis of systematic effects in the operation or inter-comparison of optical clocks, such a multi-branch approach bears the risk of adding phase noise on short and long timescales on the generated comb lines, because common-mode suppression is less effective and may depend on the specific set-up.

To circumvent such limitations, we introduce an innovative scheme for Doppler phase drift compensation, which we name dichroic heterodyne detection. As depicted in Fig. 1, a reference laser frequency  $\nu_{\text{ref}}$  here derived from a USL, is used to probe the phase difference ( $\tilde{\varphi}_1 - \varphi_1$ ) of a specific comb mode with frequency  $\nu_1$  passing through two distinct optical paths. The tilde indicates quantities where relevant phase noise is accumulated.

In our implementation, the referencing path is a short passive fibre, while the nonlinear path comprises a fibre-based optical unit with length of up to several metres in order to generate comb lines  $\tilde{\nu}_2$  at a spectral region of interest via  $\chi^{(3)}$  processes. Considering the necessity to amplify the oscillator's limited average power to produce these nonlinear effects in highly nonlinear fibres, additional phase noise and drifts may occur in the fibre chain. We show the possibility of correcting these additive phase drifts ( $\tilde{\varphi}_2 - \varphi_2$ ) occurring on  $\tilde{\nu}_2$  in the nonlinear path by detecting the phase drift present on  $\tilde{\nu}_1$  along that same path. Any Doppler contribution, that is, a phase drift corresponding to an effective optical length variation of  $x_2(t) - x_1(t)$ , is then compensated through a linear correction to obtain a Doppler-compensated phase information  $\varphi_2 = \tilde{\varphi}_2 - r(\tilde{\varphi}_1 - \varphi_1)$ , where  $r$  is, up to small dispersive effects, the ratio of the optical frequencies in the two heterodyne detections.

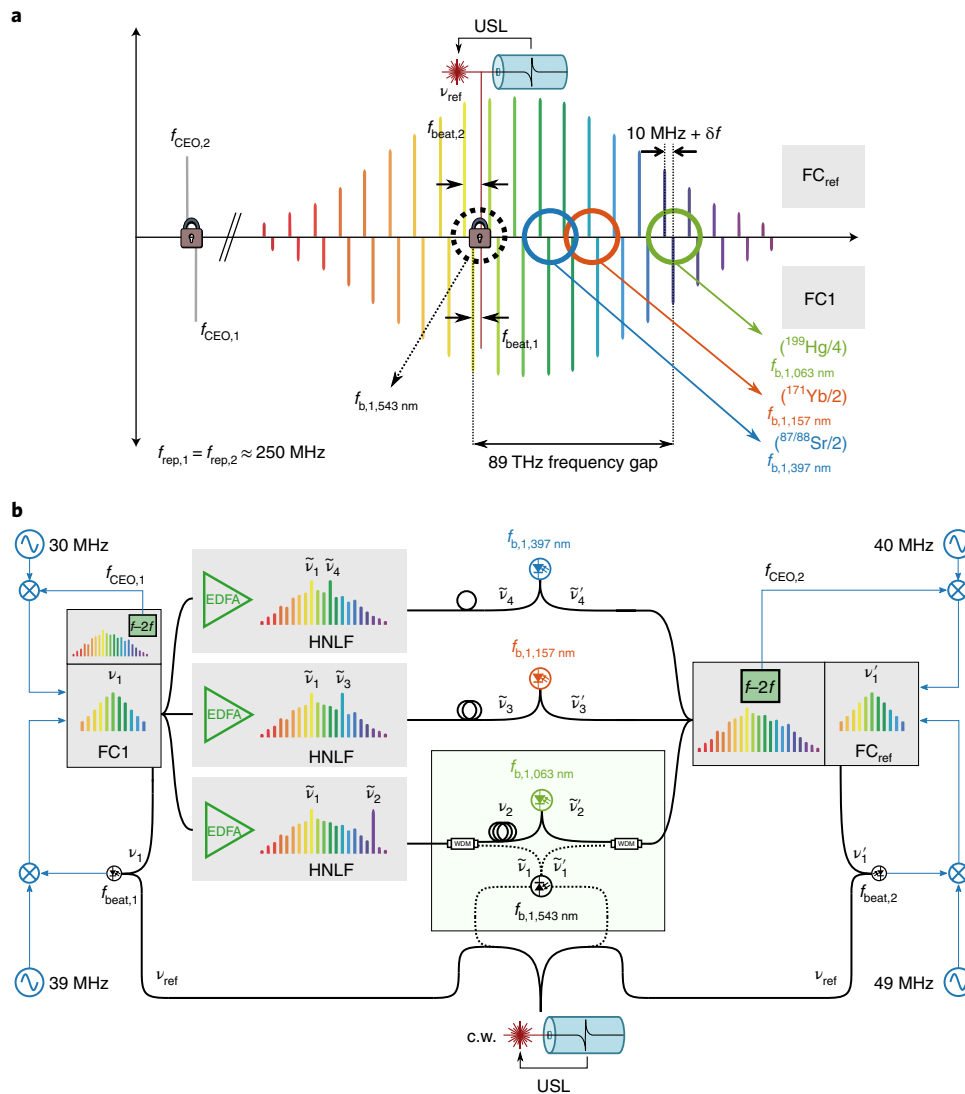
A demonstration of the dichroic heterodyne detection scheme is carried out by comparing a multi-branch FC system with a single-branch reference FC, where, at the expense of lower optical power per comb mode at the spectral regions of interest, common-mode rejection of phase noise and drifts is more effective.

## Multi-spectral comb analysis

We first show the uncompensated phase evolution of three different optical frequencies produced in three distinct paths, as extracted from multi-line heterodyne beat notes with light from the reference comb system. The corresponding spectral regions analysed comprise sub-harmonics of the narrow clock transition frequencies of neutral strontium ( $\sim 215$  THz), ytterbium ( $\sim 259$  THz) and mercury ( $\sim 282$  THz) optical lattice clocks, respectively.

As Fig. 2a shows, the two combs share the same repetition rate and are offset from one another by 10 MHz. This is achieved by a high-bandwidth phase lock ( $> 1.5$  MHz) of the individual carrier-envelope offset (CEO) frequencies with a difference of 10 MHz. In addition, the tooth  $n = 777,445$  of each comb is phase-locked to a common USL, again with a beat note difference of 10 MHz. The three multi-line beat notes are simultaneously generated at the aforementioned spectral regions and compared to the target offset of 10 MHz (see Methods, 'Heterodyne detection and phase-locked loops'). The optical phase difference between the FCs is therefore directly measurable via a high-resolution phase meter, which compares the phases of the heterodyne beat notes with the phase of the target frequency.

Figure 2b offers an overview of the measurement set-up. Two FC generators, both comprising a fibre-based mode-locked laser and a compact pre-amplifier with six output ports, are stabilized to a USL and compared at three different spectral regions along three different paths (see Methods for the details of the comb). In the multi-branch set-up, shown on the left-hand side, each spectral component is generated in a dedicated section consisting of an erbium-doped fibre amplifier (EDFA) with subsequent highly nonlinear fibre (HNLF) for spectral broadening. For the second comb, depicted on the right-hand side, all spectral components of interest are generated in a single branch, thus offering common-mode rejection of phase drifts. In this experiment, the 1,543 nm component for the reference beat note is derived from the preamplifier output, as is done for the multi-branch set-up. For the third branch, which is dedicated to the 1,063 nm spectral component, two fibre-based wavelength division multiplexers (WDMs) offer the possibility to derive the auxiliary heterodyne beat at 1,543 nm for the dichroic heterodyne detection scheme and subsequent phase drift compensation.

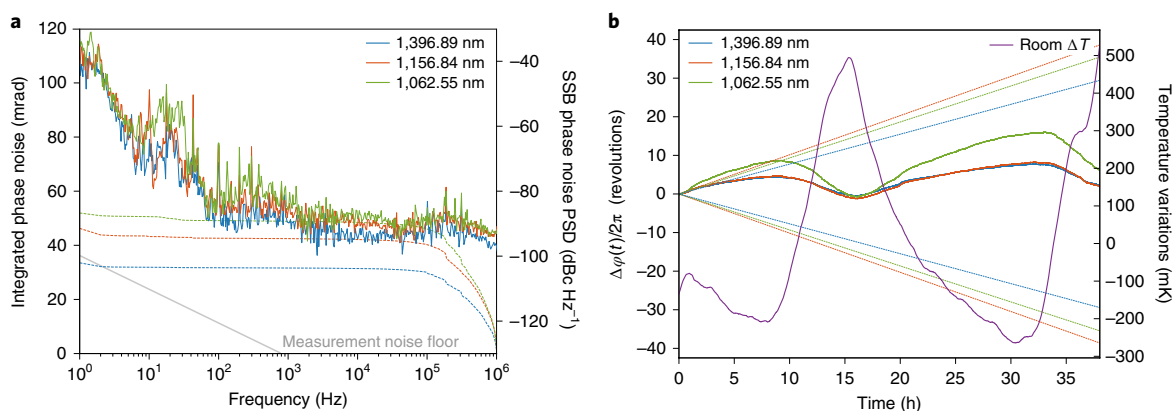


**Fig. 2 | Experimental set-up. a**, Two optical FCs (FC1 and FC<sub>ref</sub>) are set to equal repetition rates of ~250 MHz with 10 MHz difference in the CEO frequency and in the referencing optical beat notes. In particular, the two FCs are stabilized to the same sub-Hz USL in the vicinity of the femtosecond laser carriers. Three heterodyne optical beat notes are simultaneously derived by appropriate broadening and spectral filtering of the comb, indicated as  $f_{b,1,397\text{ nm}}$ ,  $f_{b,1,157\text{ nm}}$  and  $f_{b,1,063\text{ nm}}$ . An auxiliary beat note  $f_{b,1,543\text{ nm}}$ , indicated with a dashed line, is used for the dichroic heterodyne detection scheme as explained in the main text. Approximately 500 comb lines contribute to each beat note. Conveniently, there is no need for two separate auxiliary beat notes with the c.w. laser. Instead, the direct beat note  $f_{b,1,543\text{ nm}}$  between the two combs serves to simultaneously compensate drifts in the connecting fibres from both FCs. The deviation  $\delta f$  of these four beats from 10 MHz is integrated over time to obtain the phase evolution. **b**, The multi-branch set-up of FC1, which may be regarded as the device under test, is depicted on the left. Each optical target frequency as well as the octave-spanning spectrum for the  $f-2f$  interferometer is generated by an individual EDFA followed by a HNLf. On the right, the single-branch optical FC<sub>ref</sub> serves as the reference system. The heterodyne beat notes are generated in up to four individual beat detection units (BDUs) comprising narrowband optical filters at the desired spectral regions.

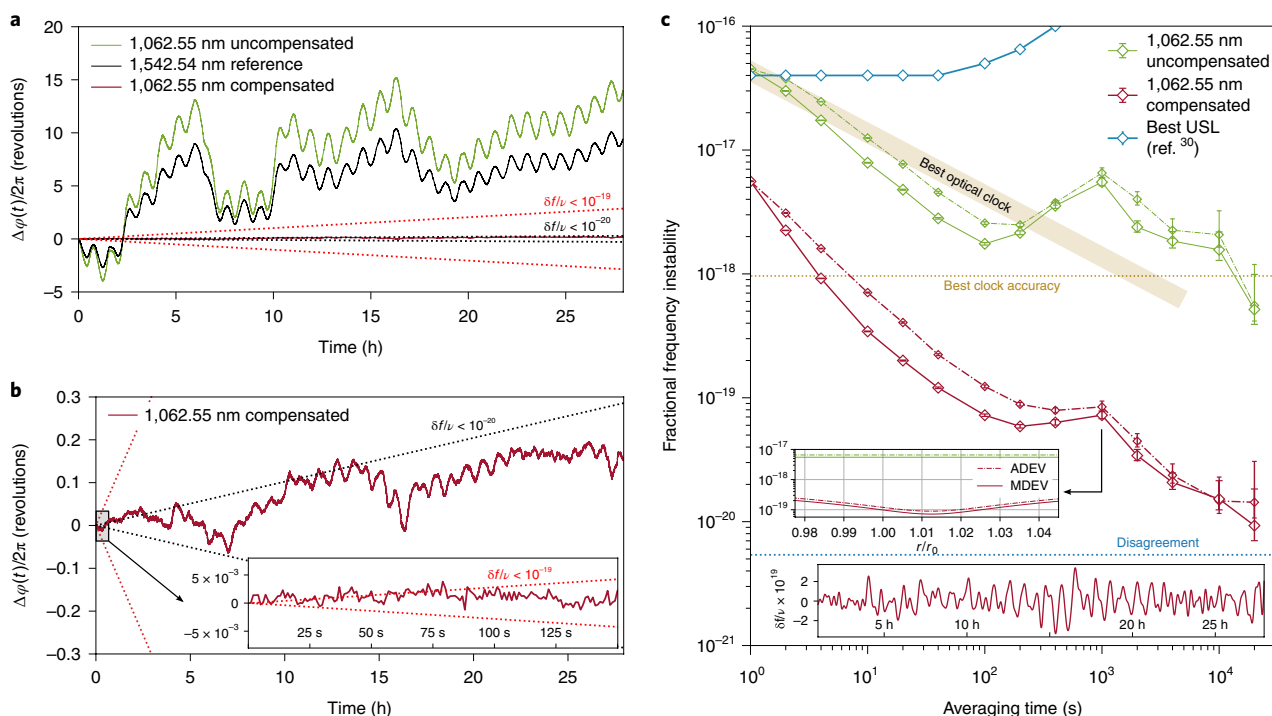
The short-term behaviour of the three heterodyne signals is measured with a phase noise analyser (see Methods). Figure 3a displays typical values of the single-sideband phase noise power spectral density (PSD, right axis) and its integration (left axis).

Figure 3b shows the temporal phase evolution of the three frequencies. Due to the known target frequency, the phase evolutions reflect the relative accuracy in the sense that the slopes correspond to frequency offsets. The dashed cones represent fractional frequency offsets of  $10^{-18}$  for the respective optical frequencies. Considering the full set of fractional frequency data, the offsets between the two combs at the end of the >35 h were  $7.63 \times 10^{-20}$ ,  $5.56 \times 10^{-20}$  and  $1.59 \times 10^{-19}$  for 1,397 nm, 1,157 nm and 1,063 nm, respectively. For all three beat notes, we find an uncertainty limited by the stability,

which is in the mid- $10^{-19}$  range. In fact, the phase tracking reveals small but relevant fluctuations, which show common correlation with the temperature evolution in the laboratory. The analysis of phase evolutions is therefore helpful in detecting possible systematic frequency shifts. In the environment where these measurements were performed, no temperature stabilization of the laboratory was present, and the resulting drifts undergo a strong day–night cycle. It is instructive to see that, in the given situation, even comparatively long measurements (for example, 10 h) may still conceal relevant frequency offsets and, therefore, measurements are more conclusive if they cover a period of at least 24 h. Even frequency data collected over many days in separate measurement campaigns may still suffer from the day–night cycle due to aliasing if the contributing data are



**Fig. 3 | Phase noise and phase tracking of the uncompensated heterodyne beat notes.** **a**, Single-sideband power spectral density (SSB PSD) of the phase noise assessed on the three heterodyne beat notes (solid lines, right axis). The dashed curves show the integrated phase noise from 1 MHz down to 1 Hz. The grey line represents the measurement noise floor of the phase noise analyser, set by its internal reference oscillator. **b**, The graphs trace the phase evolution over the course of tens of hours, revealing cycle-slip-free comb operation. The accumulated revolutions for the three beat notes, that is, the phase in units of  $2\pi$ , are scaled on the left axis. A prominent correlation between the long-term phase traces and the temperature fluctuations of the environment (in purple, right axis) is visible. The dashed lines enclose the regions of fractional frequency offsets below  $10^{-18}$ , computed for the three optical frequencies.



**Fig. 4 | Phase tracking and fractional frequency instability employing dichroic heterodyne detection.** **a**, Phase traces of the 1,063 nm (green) and auxiliary 1,543 nm (black) spectral comparisons and the resulting Doppler-compensated 1,063 nm phase evolution (red). **b**, Close-up of the Doppler-compensated phase evolution; the dashed lines enclose fractional frequency offsets in the  $10^{-20}$  and  $10^{-21}$  ranges. Inset: Magnification of the first 150 s. **c**, Fractional frequency instability in terms of the Allan deviation (ADEV, dash-dotted line) and modified Allan deviation (MDEV, continuous line) for the uncompensated (green) and Doppler-compensated (red) 1,063 nm beat notes. Error bars indicate  $1\sigma$  confidence intervals. Top inset: Instability at 1,000 s for a beat note compensated with a different scaling factor (red) and the value obtained without compensation (green). A relative scaling factor of 1 corresponds to the one that reflects exactly the nominal centre frequencies of the Bragg gratings in the beat detection units. The blue trace illustrates the lowest measured frequency instability of an USL reported so far (data from ref. 30). The shaded brown area represents the current state-of-the-art fractional frequency instability of an optical clock<sup>32</sup>. As an indication, the dashed brown line marks the lowest reported relative fractional accuracy of an optical frequency standard<sup>27</sup>, while the blue dashed line marks the relative deviation between the two combs over the whole measurement time, corresponding to  $(5.4 \pm 5.3) \times 10^{-21}$ . Bottom inset: Fractional frequency evolution over the course of the measurement time. For the latter, the data are digitally low-pass-filtered with 1 mHz, de facto applying a 500 s gate time for the frequency counter, to distinguish the low-frequency contribution from the white and flicker phase noise background.

not equally distributed over the full day. There is some similarity to the Dick effect<sup>45</sup> here, well-known for short timescales, where repetitive probing with constant period enhances noise contributions occurring at the sampling frequency and its harmonics. Individual measurements of less than 24 h may tend to always be held during similar times of the day such that there is a prominent risk of systematic frequency error.

Phase drifts may, in principle, be compensated by Doppler correction schemes involving a back-reflected signal<sup>46</sup>. However, this is often impracticable due to non-reciprocal devices present in fibre amplifier chains, such as optical isolators and frequency doublers.

### Comb analysis employing dichroic heterodyne detection

We finally apply dichroic heterodyne detection for a passive compensation scheme during a continuous measurement of the 1,063 nm branch, for a time duration exceeding 24 h. Figure 4a displays the real-time phase tracking of the 1,063 nm and 1,543 nm components along the same path. The prominent oscillation with 45 min period stems from the air-conditioning system and reveals the scaling factor  $r$  between the two phases. The trace of the 1,063 nm signal, Doppler-compensated through dichroic heterodyne detection with this factor  $r$ , is so close to zero that it is displayed in Fig. 4b, with an inset showing the rapid evolution towards the 20th digit relative uncertainty. The Allan deviation and the modified Allan deviation of both the uncompensated and the Doppler-compensated 1,063 nm signal are shown in Fig. 4c. We observe that the compensation method rejects long-term fluctuations by nearly two orders of magnitude and improves the stability and accuracy by the same amount. Short-term fluctuations on the 1 s timescale are suppressed by nearly one order of magnitude. As a result, fractional frequency instability in the  $10^{-20}$  range is reached within 100 s, averaging down to  $1.5 \times 10^{-20}$  after only 10,000 s. On the left side of Fig. 4c we show an inset displaying the instability at 1,000 s for the compensated beat note as a function of the relative scaling factor  $r/r_0$  (red), with the uncompensated instability given as reference (green). A value of 1 corresponds to a scaling factor  $r$  equal to the ratio  $r_0$  of the nominal centre frequencies of the Bragg gratings in the beat detection units. We picked the timescale of 1,000 s because of the high temperature fluctuations in such period, and we used the observed phase fluctuations of the comb–comb comparison to calibrate the compensation. On the uncompensated fractional frequency trace, the relative offset between the two combs corresponds to  $(4.92 \pm 3.35) \times 10^{-19}$ , whereas on applying the dichroic compensation we find a relative deviation between the two combs of  $(5.4 \pm 5.3) \times 10^{-21}$  at 1,063 nm over the full 28 h dataset. In the lower inset of Fig. 4c, we show the frequency trace of the compensated data. A digital 1 mHz low-pass filter was used to visualize the remaining fluctuations otherwise covered by the short-term white phase noise.

### Discussion

The level of comb fidelity demonstrated here supports sub-mHz stability for wide-band optical frequency synthesis. For an ultimate determination of the level of accuracy that such comb technology can achieve, further decorrelation in the qualification might be desirable, such as comb operation in separate rooms or the use of different comb technologies.

In conclusion, real-time optical phase tracking serves as an essential tool for resolving subtle phase drifts that would degrade the accuracy for optical frequency measurements. Moreover, the technique of dichroic heterodyne detection offers phase drift compensation, which is becoming highly relevant in situations where compensation schemes with counter-propagating beams are impractical. For FCs where separate arms are used to achieve optimized spectral power and maximized heterodyne beat notes' signal-to-noise ratios, this technique can provide relative accuracy at the low  $10^{-20}$  level. The comb technology involved in this work has

shown phase-locked loops with cycle-slip-free operation over days, and the resulting performance paired with the demonstrated robustness alleviates the need for transfer oscillator schemes for frequency measurements even at the level of  $10^{-20}$ . The demonstrated system thus represents a powerful enabling technology in the domain of optical frequency standard comparisons and clockworks in future optical atomic clocks, which are now under serious investigation for a potential re-definition of the International Time Scale<sup>47–51</sup>.

### Online content

Any methods, additional references, Nature Research reporting summaries, source data, statements of code and data availability and associated accession codes are available at <https://doi.org/10.1038/s41566-019-0520-5>.

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## Author contributions

M.G. and W.H. performed the experiments, conceived and realized the two frequency combs and the optical set-up. M.G. analysed the data. M.G. and W.H. wrote the manuscript. M.F. and M.L. managed the project. T.U. provided deep insight into the interpretation of the results and aided with the optical set-up. R.H. initiated and led the activities. All co-authors commented on and improved the manuscript.

## Competing interests

The authors declare no competing interests.

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## Methods

**Ultra-low-noise optical FC.** The core of the ultra-low-noise FC is a passively mode-locked erbium-doped fibre laser oscillator emitting sub-100 fs pulses centred at a wavelength of 1,560 nm. Mode locking is achieved through an artificial saturable absorber realized as an all polarization-maintaining (PM) nonlinear amplifying loop mirror<sup>52</sup>. The repetition rate of 250 MHz is coarsely tunable by  $\pm 2.5$  MHz through motorized adjustment of a reflecting mirror, and it is further adjustable through a piezo-actuator, both acting on the laser cavity length. For fast simultaneous stabilization of a specific comb mode to a reference laser and of the CEO frequency to a radio-frequency reference, two orthogonally acting electro-optic modulators are implemented inside the laser cavity. The first modulator acts in the traditional way, stretching the comb modes around a fix point close to zero optical frequency, whereas the second one stretches the modes around a fix point at  $\sim 191$  THz (1,560 nm), close to the carrier frequency<sup>53</sup>.

The oscillator output is preamplified for both FCs to  $\sim 100$  mW of output power, offering an  $\sim 60$ -nm-wide spectrum around 1,560 nm. This light is distributed to six output ports using cascaded  $1 \times 2$  and  $1 \times 3$  fused fibre couplers, allowing for  $\sim 60$  fs pulse duration if compressed in standard PM fibre. For each FC, one port is amplified, spectrally broadened and subsequently used for detection of the CEO frequency in an  $f$ - $2f$  interferometer. Another port in each FC is dedicated to the generation of the optical beat with the cavity stabilized c.w. laser. In the case of the multi-branch set-up, three further output ports seed the amplifier units for the generation of comb lines in the spectral regions of interest, described in the 'Amplifier branches' section below. The mode-lock conditions and laser parameters stay basically unchanged during operation, and no spontaneous loss of mode locking occurred while operating the FCs.

**Amplifier branches.** The branches of FC1 and the two  $f$ - $2f$  interferometer modules share a similar architecture consisting of  $\sim 7$  m of PM fibre. The preamplifier output is further amplified in a forward and backward pumped EDFA and spectrally broadened in a PM HNLF. In the case of the multi-branch set-up, the power of the amplifier pump diodes is tuned individually to optimize the optical PSD in the region of interest. In the case of the single-branch set-up, a compromise with respect to pump powers has to be found while aiming to optimize the CEO frequency signal and the optical power at the target frequencies.

**Optical reference system.** As a common optical reference for the FC comparison, a 1,542 nm distributed-feedback fibre laser (E15 Koheras BASIK, NKT Photonics A/S) is stabilized to a high-finesse Fabry–Pérot cavity made from ultra-low-expansion glass<sup>54</sup>. The cavity features a free spectral range of 1.2 GHz with a finesse of roughly 400,000. Direct feedback to the laser's resonator length is accomplished through an internal piezo actuator with a bandwidth of 10 kHz, while additional feedback through an acousto-optic modulator between laser and cavity increases the overall locking bandwidth to 300 kHz. The resulting fractional frequency stability given in terms of Allan deviation corresponds to  $1.5 \times 10^{-15}$  at 1 s comparing two identical systems, assessed with a II-mode frequency counter, 100 ms gate time and 1 MHz measurement bandwidth. The residual drift with respect to a GPS reference is on the order of  $150 \text{ MHz s}^{-1}$ , dominated by ageing of the glass.

**Heterodyne detection and phase-locked loops.** There are two established methods to assess a comb's relative instability, as described in ref. <sup>37</sup>. The first makes use of a transfer laser generating beat notes with a close-by comb line, while the second method uses direct multi-line heterodyne detection of the two FCs in a narrow spectral window. The latter implies the necessity to operate the two FCs at the same repetition rate and to arrange simultaneous arrival of the pulses on the photodetector. Typically, this renders the system susceptible to loss of pulse overlap during extended measurement times. In this work, days of cycle-slip-free, uninterrupted and phase-coherent operation of the FCs permits us to make use of the second method.

All fibre links for the detection of heterodyne signals are accomplished through standard PM fibre without active length stabilization. The links have typical lengths between 1 m and 2 m, residing on or in close proximity to the water-cooled base plate. They have been carefully shielded from vibrations and air currents, and a good thermal shield is applied over the set-up by using padding material and temperature-isolating foil (Mylar foil). We note that the amplifiers and connecting fibres of the first and second system have not been explicitly tuned to the same length. However, due to the similarity of the amplifiers, the length difference is at

most on the order of 2 m, offering some common-mode rejection of temperature drifts in the rather long fibre amplifiers.

For all heterodyne detections, that is, for the optical referencing and for the comparison of the combs, the two engaged light fields are superimposed in a fused fibre coupler with subsequent passage through a 100 GHz wide optical filter. For the multi-line heterodyne detection, this involves  $\sim 500$  comb lines.

Both FCs are stabilized in their  $n$ th comb mode to the reference laser, to yield  $f_{\text{cw}} = f_{\text{CEO},i} + n f_{\text{rep},i} + f_{\text{beat},i}$ , where  $n$  is 777,445 and the index  $i$  refers to the respective FC set-up. The two CEO frequencies  $f_{\text{CEO},i}$  are phase-locked to  $-30$  MHz and  $-40$  MHz, and the optical reference beat notes  $f_{\text{beat},i}$  are 39 MHz and 49 MHz, respectively. If all systems were perfect, this would lead to exactly the same repetition rate for both combs. All necessary RF frequencies are derived through direct digital synthesizers, in turn referenced to a common low-noise 10 MHz oscillator (Timetech REFGEN). In this setting, any drift or noise of the reference laser is cancelled if the FCs stay phase-locked.

For the multi-line comparison on a first path, the necessary pulse overlap is achieved by initial detuning of the repetition rate of one FC with respect to the other. Having a fully fibre-based set-up without adjustable delay line, any further heterodyne beat note requires precise length tuning when splicing the fibre connections. This is done iteratively by observing the pulse spacing at the new heterodyne signal when pulse overlap at the first path is achieved, and by subsequent correction of the fibre length. Despite the fact that a compromise has to be found for simultaneous pulse overlap for the three heterodyne beats as well as for the spectral broadening in the single-branch set-up, the signal-to-noise ratio for any of the signals is above 45 dB at 100 kHz resolution bandwidth, corresponding to a white phase noise plateau of at most  $-95 \text{ dBc Hz}^{-1}$ . The single-sided power spectral density was measured by an externally referenced phase noise analyser from Keysight (PXA N9040A), which compares the signal at the input with an internal low-noise oven-controlled quartz oscillator featuring a noise floor well below the PSD levels shown in Fig. 3.

For high-resolution frequency and phase tracking of the beat notes, a dead-time free frequency counter was used (FXE, K+K Messtechnik<sup>55</sup>). This is based on a synchronously sampling multichannel progressive phase recorder with internal time base of 1 ms. The data were recorded with a gate time of 1 s, and the input channels were externally band-pass-filtered at 10 MHz with a 3 dB bandwidth of  $\sim 4$  MHz.

**FC uncertainty.** Assessment of the FC uncertainty is carried out by analysing the offset of the heterodyne beat notes with respect to the 10 MHz target frequency. The latter is defined by the four direct digital synthesizer reference frequencies setting the combs, commonly phase-locked to the 10 MHz signal used to reference the frequency counters and the phase noise analyser. In this way, the phase of the 10 MHz radio-frequency directly reveals the phase of the optical-frequency components generating the beat signal. We have chosen to show the full-time evolution of this phase in the main text to give full insight into the system dynamics. To characterize the uncertainty of the total measurement by a single number, we give the value of the modified Allan deviation at an averaging time corresponding to a quarter of the total acquisition time, that is,  $\delta f = \sigma_{\text{mod}}(\tau = T/4)$ . No extrapolation is done towards the total measurement time  $T$ .

## Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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